BRD

BUSINESS REQUIREMENTS DOCUMENT

Employee Onboarding Dashboard

Field	Value
Document ID	BRD-EOD-001
Version	1.0
Date	[Date]
Author	[Author]
Status	Draft — pending stakeholder review
Scope of this document	End-to-end business requirements for Phases 0 through 6 of the Employee Onboarding Dashboard
Target platforms	Responsive Web Application (Chrome, Edge, Firefox, Safari down to 768px viewport)
Reference	[Reference/supersedes — none for v1.0]

Top Five Features

#	Feature	The question it answers	Section
1	AI-Assisted Task Drafting with Human Approval	"How can a manager generate a tailored 30-day task plan from a short brief in seconds without risking unreviewed AI content reaching the new hire?"	Section 4.2
2	Unified Joinee Onboarding Dashboard	"What exact tasks, mandatory certifications, and key introductory contacts must I complete this week, and what is currently overdue?"	Section 4.4
3	Real-Time Manager Tracking Dashboard	"Which of my new hires are behind schedule on compliance or tasks right now without me having to ask them for a status update?"	Section 4.4
4	Static Reference Information Hub	"Where can a new hire find authoritative instructions for IT access, tool requests, and company policies without submitting repetitive support tickets?"	Section 4.4

#	Feature	The question it answers	Section
5	Compliance & Certification Management	"How do we verify that every mandatory compliance certification is assigned, tracked against hard deadlines, and backed by uploaded proof?"	Section 4.3

1. Purpose

New joinees currently receive onboarding instructions across fragmented emails, ad-hoc Slack messages, and verbal handoffs. The organization lacks a single system of record that outlines what tasks must be completed, by when, which certifications are mandatory, whom to meet, or how to navigate IT and HR access requests. Consequently, managers rebuild identical task lists from scratch for every hire, ramp-up velocity varies significantly across teams, mandatory compliance deadlines are missed, and IT and HR support channels are congested with repetitive baseline inquiries.

This document defines the business requirements for the Employee Onboarding Dashboard (EOD), a lightweight internal web platform with two explicit roles: Manager and New Joinee. The application centralizes onboarding plans, compliance certifications, introductory stakeholder connections, and an authoritative, searchable reference hub.

Artificial intelligence is constrained to a single capability: expanding a manager's unstructured brief into a structured, draft task schedule that requires mandatory human review and approval prior to publication. This document sets the functional, technical, and governance boundaries for delivery.

2. Objectives

ID	Objective	Success Measure
OBJ-01	Give every new joinee a single dashboard covering their tasks, certifications, contacts and support information from day one	100% of new joinees have an approved, active onboarding plan visible on or before their start date

ID	Objective	Success Measure
OBJ-02	Cut the manager effort required to prepare an onboarding plan	Manager time to produce and assign a complete plan reduced from [TBD - baseline to confirm] to under 20 minutes per hire
OBJ-03	Make onboarding progress visible to managers without manual follow-up	Manager can see per-joinee task, certification and connection completion status without messaging the joinee; 0 manual status-check requests needed
OBJ-04	Ensure mandatory certifications are completed on time	95% of mandatory certifications completed by their stated deadline
OBJ-05	Reduce repetitive access and support queries reaching IT and HR	Measurable reduction in first-30-day tickets and emails asking where to find access, tools or policy information [TBD - confirm baseline with IT]
OBJ-06	Produce AI task drafts that managers find usable rather than rewrite wholesale	AI draft acceptance rate above 70%, with fewer than 30% of generated tasks edited or deleted before assignment
OBJ-07	Standardise onboarding quality so it does not depend on which manager a joinee reports to	Variance in 30-day checklist completion rate across managers within [TBD] percentage points

3. Scope

3.1 In scope

- Two roles only: `` `MANAGER` `` and `` `JOINEE` ``, with strict manager-to-joinee data access mapping.
- Authentication via corporate OpenID Connect (OIDC) identity provider (Google Workspace / Azure AD).
- Onboarding plan creation per joinee, anchored dynamically to the joinee's start date.
- AI-assisted task generation: expanding manager free-text briefs into structured draft task schedules.
- Mandatory manager review, editing, and approval workflow for all AI-generated drafts prior to joinee publication.

- First-class manual task creation, editing, and deletion independent of the AI engine.
- Manual entry and tracking of role certifications (name, provider, URL, deadline, mandatory flag).
- Manual entry and tracking of introductory connections (name, email, job title, rationale).
- Joinee dashboard featuring weekly task breakdowns, overdue item highlighting, and completion progress indicators.
- Joinee task state updates (To Do, In Progress, Done, Blocked), certification state tracking, and certification proof document uploads.
- Joinee introductory connection status tracking (Pending, Scheduled, Met).
- Read-only Static Information Hub categorized across IT/Access, Accounts/Tools, HR/Payroll, Workplace, Security/Compliance, and FAQ.
- Versioned database seed file ingestion for Information Hub updates without code redeployment.
- Full keyword search across Information Hub titles and Markdown body text.
- Automated task due-date derivation based on joinee start date plus task day offset.
- Automated email reminder dispatches for approaching and overdue deadlines, with manager escalation.
- Immutable audit logging for plan approvals, task assignments, and completion events.
- Comprehensive logging of every AI generation request (inputs, outputs, latency, tokens, acceptance, edit states).

3.2 Explicitly deferred

The following items are deferred from v1.0 and represent candidates for subsequent delivery phases (detailed in Section 10):

- **AI-generated recommendations for introductory connections**: Requires integration with an authoritative corporate directory and introduces organizational bias concerns; manual manager entry is reliable on day one.
- **AI-written progress summaries for managers**: Adds operational LLM cost and review overhead for negligible gain over deterministic progress metrics.
- **Retrieval-Augmented Generation (RAG) assistant for document queries**: Requires dedicated vector infrastructure, corpus maintenance, and document-level permission controls; the static Information Hub resolves standard questions at lower complexity.

- **AI-suggested role certifications**: Compliance and regulatory certifications require absolute accuracy; automated suggestions present unacceptable compliance risk.
- **Automated HRIS synchronization**: Initial hire volume supports manual and CSV provisioning; avoids premature vendor API coupling [TBD - confirm HRIS vendor].
- **Slack and Calendar integration**: Transactional email handles reminders sufficiently in v1 without adding third-party API dependencies.
- **Reusable role-based plan templates**: Deferred until empirical usage patterns identify recurring cross-team task sets.
- **Dedicated HR administrator role**: The strict two-role model prevents RBAC complexity; HR updates content via database seed migrations.
- **Native mobile applications**: A responsive web interface fulfills all requirements for desk-based new hires during their first 30 days.
- **Cross-organizational onboarding analytics**: Meaningful benchmarking requires data accumulated over multiple completed hiring cycles.

3.3 Permanently excluded (not a phasing decision)

The following capabilities are permanently excluded from the system architecture and will not be implemented in future phases:

- Any AI application beyond task-list drafting (certifications, people recommendations, Information Hub content, and progress summaries will never be AI-generated).
- Direct publishing of AI-generated content to joinees without explicit, manual manager approval.
- Performance management, employee appraisal, goal-setting, or rating functionality.
- Payroll execution, compensation recording, or benefits enrollment.
- Applicant tracking, recruitment workflows, or pre-offer candidate tracking.
- Document e-signature or formal employment contract execution.
- Learning Management System (LMS) capabilities; the system provides outbound hyperlinks to courseware but hosts no learning modules.
- Serving as the primary system of record for corporate employee master data.
- Joinee-to-joinee social feeds, open messaging, or direct chat functionality.

- Any public, external, or non-authenticated network access.

4. Detailed Requirements

4.1 Authentication, Access Control & User Administration

ID	Requirement	Priority
BR-01	The system shall authenticate all users through the corporate OIDC identity provider and assign exactly one role, MANAGER or JOINEE, per user.	M
BR-02	The system shall restrict a manager to viewing and modifying only the joinees mapped to them, rejecting all other access with an authorisation error.	M

4.2 Onboarding Plan Lifecycle & AI Task Generation

ID	Requirement	Priority
BR-03	The system shall allow a manager to create an onboarding plan for a joinee, capturing start date and plan duration, with the plan created in DRAFT status.	M
BR-04	The system shall accept a free-text brief from the manager and send it, together with the joinee's job title, team, start date and plan duration, to the AI task generation service.	M
BR-05	The system shall require the AI service to return tasks as structured JSON conforming to a defined schema, with each task carrying a title, description, category, week number, day offset, priority and optional subtasks.	M
BR-06	The system shall validate every AI response against the defined JSON schema server-side and reject any non-conforming response.	M
BR-07	The system shall retry a failed or invalid AI generation once, and on a second failure present the manual task entry form without loss of any data the manager has already entered.	M
BR-08	The system shall limit a single AI generation to a maximum of 15 tasks.	S
BR-09	The system shall present all AI-generated tasks in an editable grid where the manager can modify any field, delete rows and add new rows before assignment.	M

ID	Requirement	Priority
BR-10	The system shall prevent any task from becoming visible to a joinee until the manager has explicitly approved and assigned the plan.	M
BR-11	The system shall allow a manager to create, edit and delete tasks manually without invoking AI at any point.	M
BR-12	The system shall calculate each task's due date as the joinee's start date plus the task's day offset at the moment of assignment.	M
BR-13	The system shall transition a plan from DRAFT to ACTIVE on manager approval and make it visible to the joinee at that point.	M
BR-29	The system shall log every AI run with the manager identity, input brief, raw output, validation result, token usage, latency, and whether the output was accepted and whether it was edited.	M
BR-31	The system shall allow a manager to mark a plan COMPLETED once all mandatory items are done or the plan period has elapsed.	C

4.3 Certifications & Introductory Connections

ID	Requirement	Priority
BR-14	The system shall allow a manager to add, edit and remove certifications for a joinee, capturing name, provider, URL, deadline and a mandatory flag.	M
BR-15	The system shall allow a manager to add, edit and remove people for the joinee to connect with, capturing name, email, job title and a reason for the connection.	M
BR-16	The system shall store connection contacts as free text rather than as references to system users, so that people outside the application can be listed.	M
BR-19	The system shall allow a joinee to update certification status and upload a proof document for a completed certification.	S
BR-20	The system shall allow a joinee to mark each listed person as Pending, Scheduled or Met, recording the date when set to Met.	M

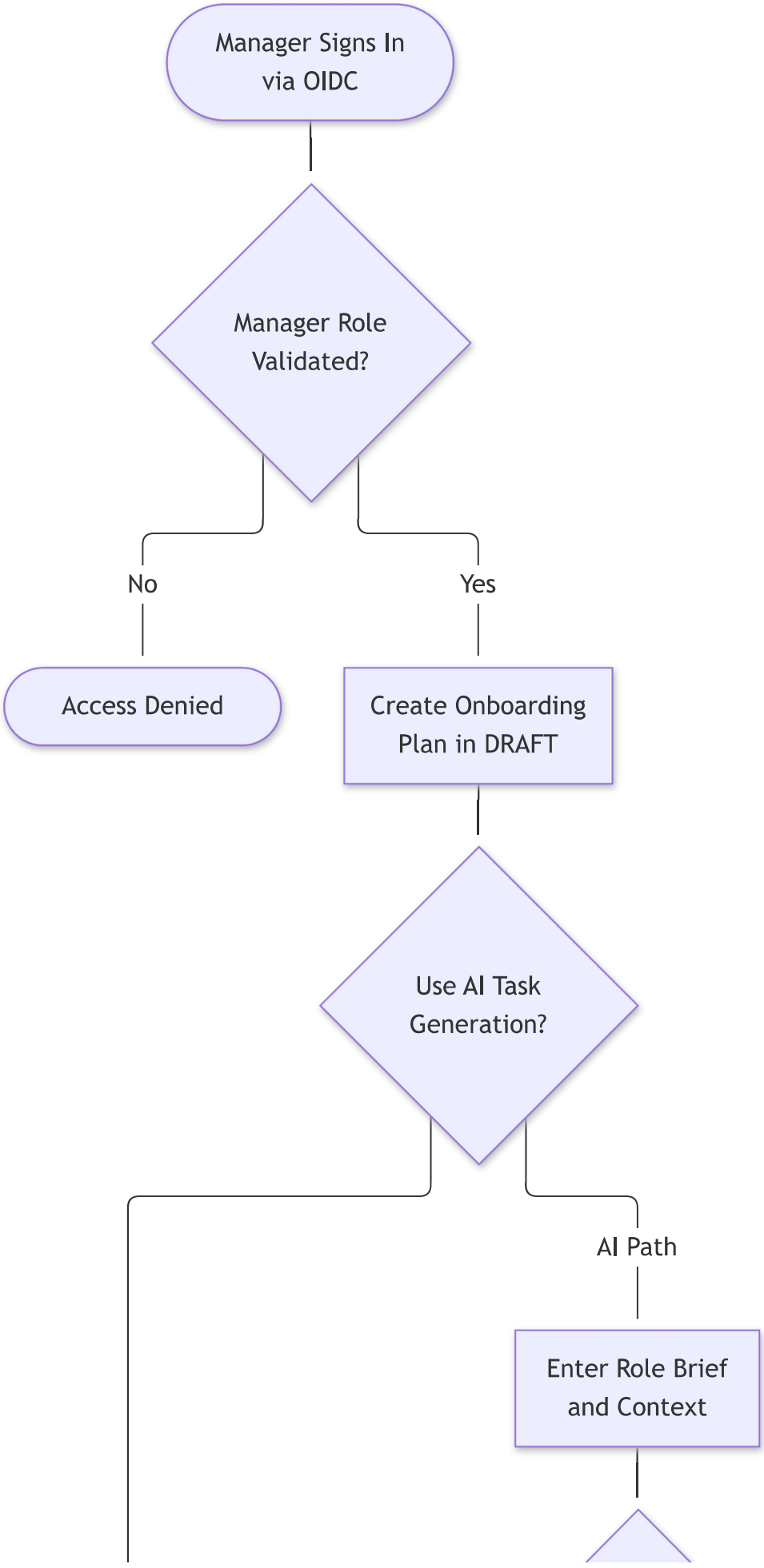
4.4 Joinee & Manager Dashboards, Information Hub & Notifications

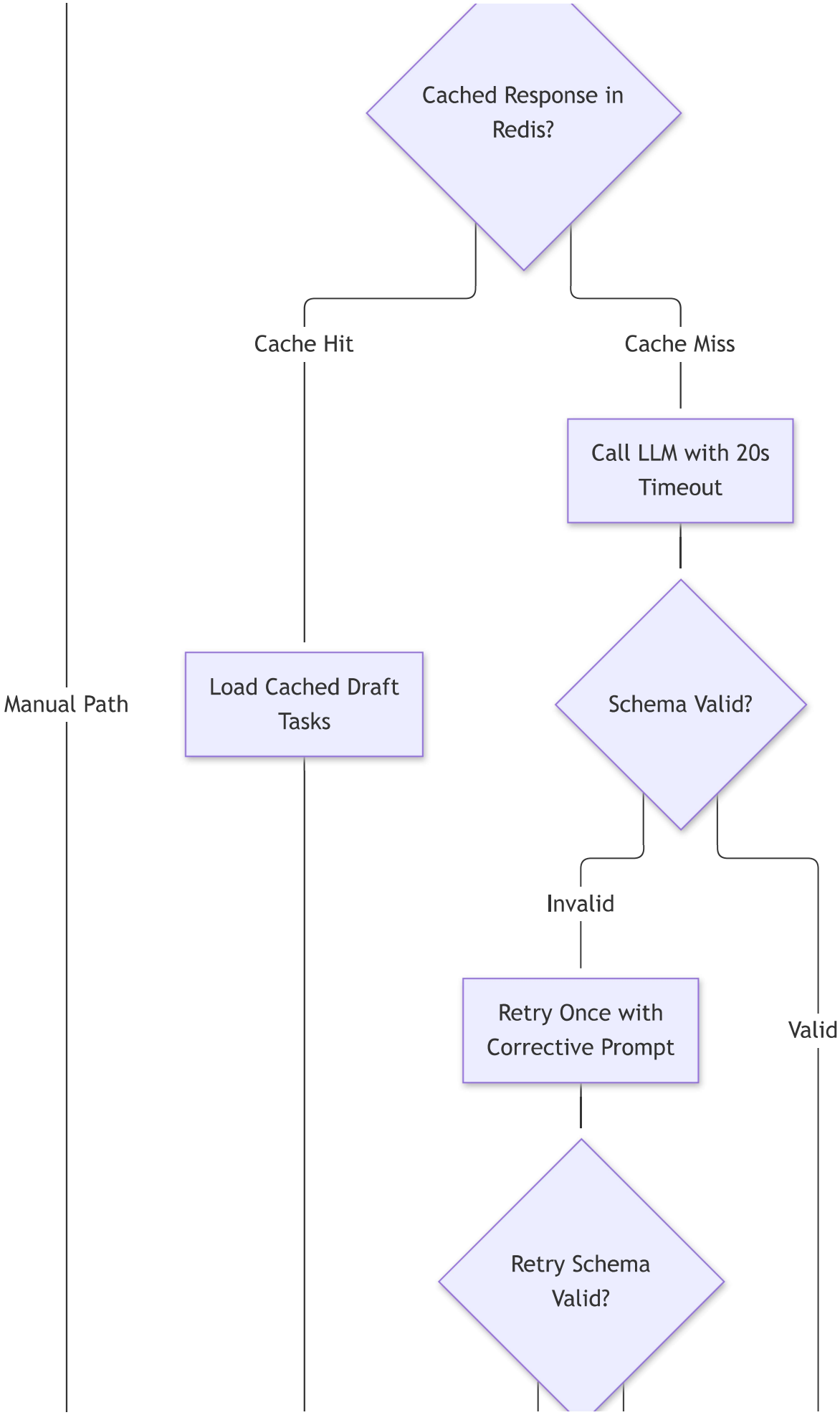
ID	Requirement	Priority
BR-17	The system shall display to the joinee their active plan grouped by week, showing task title, description, category, due date, priority and current status.	M
BR-18	The system shall allow a joinee to change a task status between To Do, In Progress, Done and Blocked, recording the completion timestamp when set to Done.	M
BR-21	The system shall display a joinee dashboard showing overall completion percentage, the current week's tasks, and all overdue items.	M
BR-22	The system shall display a manager dashboard listing every joinee mapped to that manager with completion percentage and an overdue-item count.	M
BR-23	The system shall present a read-only Information Hub to joinees, organised by category, covering IT helpdesk and ticket raising, access requests, accounts and tools, HR and payroll, workplace policies, security and compliance, and a general FAQ.	M
BR-24	The system shall load Information Hub content from a versioned seed file into the database at deployment, so that content can be corrected without a code change.	M
BR-25	The system shall provide a keyword search across Information Hub entry titles and body content.	S
BR-26	The system shall allow an Information Hub entry to be deactivated so it is hidden from joinees without being deleted.	C
BR-27	The system shall send the joinee an email reminder for tasks and certifications approaching their due date.	S
BR-28	The system shall escalate overdue items to the assigning manager by email.	S
BR-30	The system shall maintain an audit trail of plan approval, task assignment, status changes and completions, recording actor and timestamp.	S
BR-32	The system shall allow a joinee to add a note or comment against a task to flag a blocker to the manager.	C

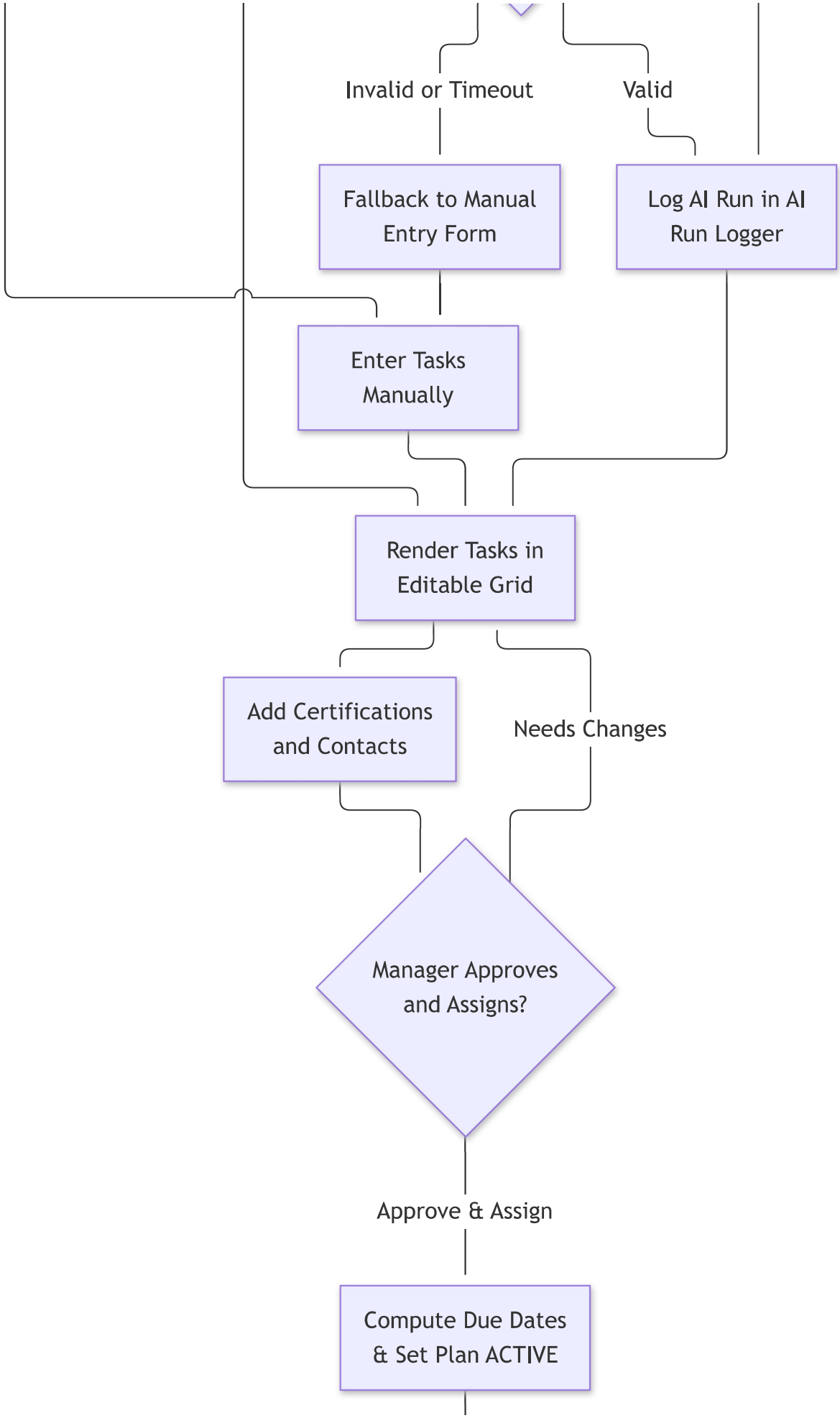
5. Solution Approach

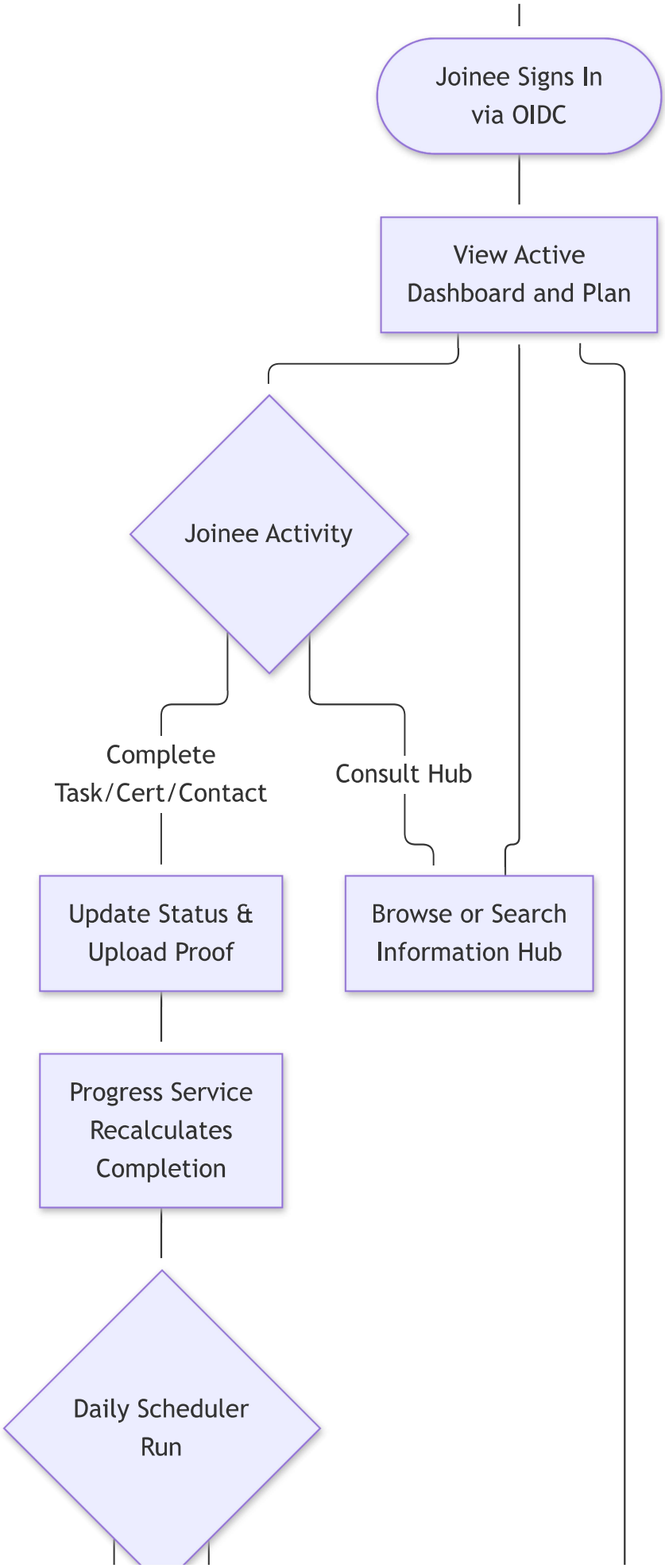
5.1 Process Flow

The onboarding lifecycle executes sequentially across authentication, plan formulation, manager governance, joinee execution, and scheduled monitoring. The manager signs in via corporate OIDC, selects a mapped joinee, and initiates a draft plan. The manager enters an unstructured brief which the system transmits alongside joinee metadata to the AI service (or checks Redis cache). The resulting task list is validated server-side against a strict JSON schema and rendered into an interactive grid for manager editing alongside manual certification and contact entries. Upon explicit manager approval, the plan transitions to active, calculating calendar due dates from start offsets and publishing them to the joinee dashboard. The joinee completes items, uploads certification proofs, and references the seeded Information Hub. Background schedulers track operational deadlines daily, triggering email reminders and manager escalations.


Manager Signs In via OIDC
Manager Role Validated?
No
Yes
Access Denied
Create Onboarding Plan in DRAFT
Use AI Task Generation?
AI Path
Enter Role Brief and Context


Cached Response in Redis?
Cache Hit
Cache Miss
Call LLM with 20s Timeout
Load Cached Draft Tasks
Schema Valid?
Manual Path
Invalid
Retry Once with Corrective Prompt
Valid
Retry Schema Valid?


Invalid or Timeout
Valid
Fallback to Manual Entry Form
Log AI Run in AI Run Logger
Enter Tasks Manually
Render Tasks in Editable Grid
Add Certifications and Contacts
Needs Changes
Manager Approves and Assigns?
Approve & Assign
Compute Due Dates & Set Plan ACTIVE


Joinee Signs In via OIDC
View Active Dashboard and Plan
Joinee Activity
Complete Task/Cert/Contact
Consult Hub
Update Status & Upload Proof
Browse or Search Information Hub
Progress Service Recalculates Completion
Daily Scheduler Run

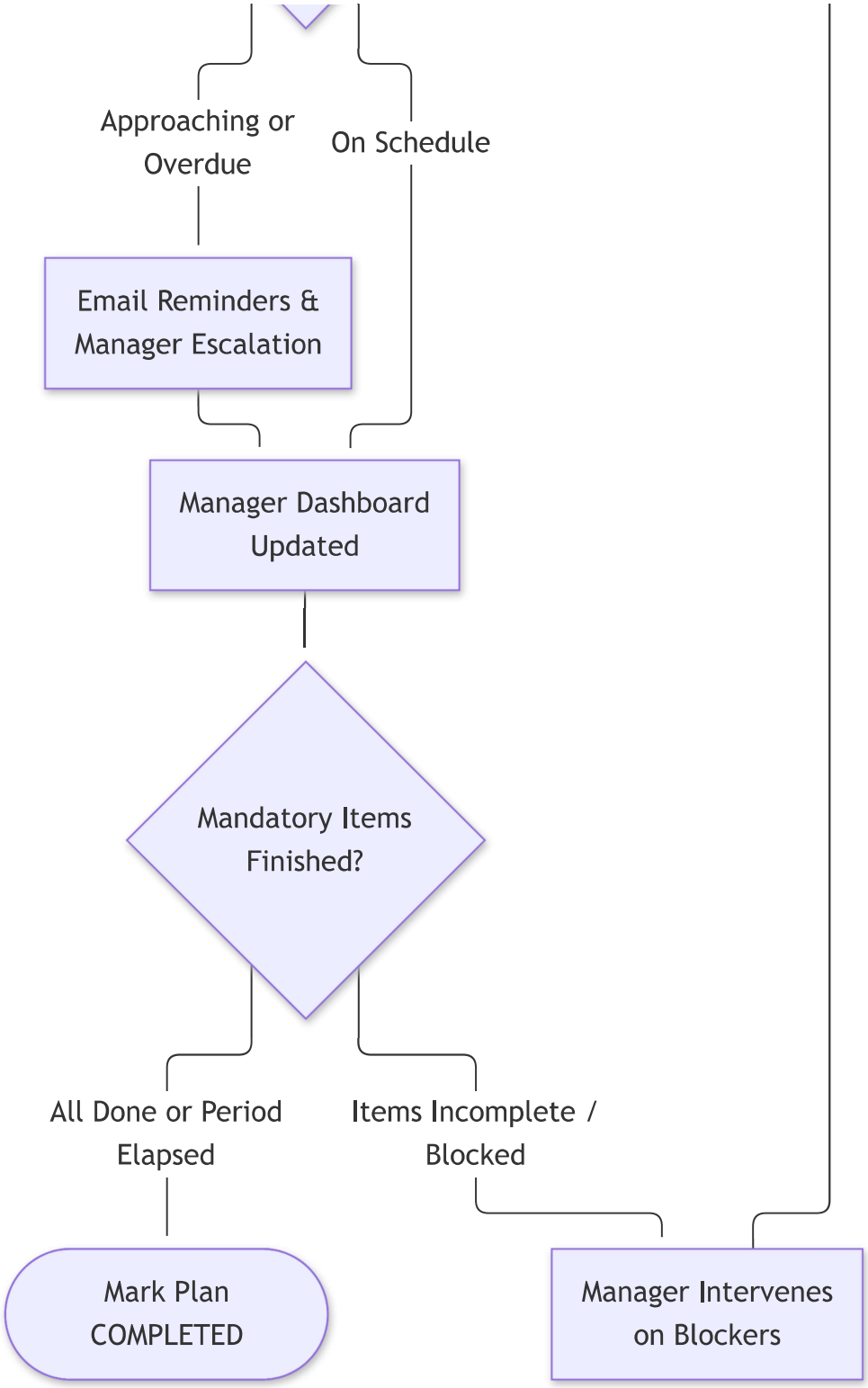


Figure 1 — Process Flow

5.2 Architecture

The platform is architected as a modular web application comprising a responsive React frontend, a Spring Boot REST API, a PostgreSQL relational datastore, Redis caching, and S3-compatible object storage. User requests terminate at an API Gateway handling TLS and routing. Core business logic is partitioned into dedicated Spring services: Auth & RBAC validates OIDC tokens and enforces data

ownership; Plan & Task coordinates plan state transitions and date calculations; AI Orchestration manages model communication, schema validation, retries, and run telemetry; Progress aggregates real-time completion statistics; and Notification orchestrates scheduled SMTP alerts. Reference data is ingested via automated database seed loaders.

Figure 2 — Architecture

5.3 Proposed Technology Stack

Layer	Proposal	Notes
Frontend Framework	React 18 with TypeScript & Vite	Provides component modularity, strict client-side typing, and fast development turnaround for grid-heavy workflows.
UI Styling	Tailwind CSS	Delivers utility-first styling ensuring responsive alignment across desktop and tablet (down to 768px).
Backend Framework	Java 21 with Spring Boot 3.x	Matches core organizational skills; provides enterprise-grade transaction management, robust validation, and native security controls.
Database	PostgreSQL 16	ACID-compliant relational storage ensuring data integrity across complex user-plan-task relational graphs.
Database Migration	Flyway	Manages immutable schema migrations and version-controlled seed data loading for the Information Hub.
In-Memory Cache	Redis	Caches AI responses via input-context hashing to prevent token waste and manages temporary session states.
Object Storage	S3-Compatible Store (MinIO / AWS S3)	Decouples binary certification proof file storage from the relational database schema.
Authentication	OIDC / OAuth2 with JWT	Leverages existing corporate directory identities (Azure AD / Google Workspace) without local credential exposure.
Scheduling	Spring Scheduler	Lightweight, in-process cron execution for daily deadline evaluation and notification dispatch.
Email Service	Transactional SMTP	Standardized corporate mail transport for reminders and overdue escalations without external SaaS dependencies.

Layer	Proposal	Notes
API Specification	OpenAPI 3.1	Enforces a strict, versioned API contract between backend services and the frontend client prior to UI construction.

5.4 Design Principles

- **Manager Accountability**: AI provides draft recommendations only; no task, deadline, or milestone reaches a joinee without explicit, audited human manager approval.
- **Deterministic Fallbacks**: External AI service outages, rate limits, or schema validation failures must never block onboarding operations; manual workflows remain fully functional at all times.
- **Zero-PII Model Transmission**: No personal identifiable information (names, emails, corporate IDs) is ever transmitted to third-party LLM providers; payloads are restricted to job titles, teams, and briefs.
- **Strict Server-Side Authorization**: Client-provided identifiers are never trusted for ownership; manager-to-joinee data access is evaluated cryptographically on every endpoint.
- **Decoupled Reference Content**: Information Hub resources are maintained independently of the application source code through database seed configurations, allowing instant operational corrections.

---

6. Non-Functional Requirements

Security

- All network traffic must use HTTPS with TLS 1.3; all API endpoints must require authentication with zero anonymous access allowed across any resource.
- Data ownership authorization must be verified server-side on every request; a manager ID in a request payload must never be accepted as proof of ownership.
- Uploaded certification proof files must be scanned for malware, restricted to a configured file-type allowlist (PDF, PNG, JPEG), and subject to strict file size limits.

Privacy

- Payloads sent to external LLM providers must contain only the minimum required operational context (job title, team, start date, plan duration, brief); no employee names or email addresses may be sent.
- Third-party LLM agreements must guarantee that submitted enterprise data is excluded from model training, with documented and approved provider data-retention terms.

Data Retention

- Completed onboarding plans must be retained for [TBD] months post-completion.
- AI generation telemetry and execution logs must be retained for [TBD] months for auditing and prompt tuning.
- Uploaded certification proof files must be retained in accordance with corporate HR compliance policy [TBD].

Auditability

- Plan approvals, task assignments, status changes, and completion timestamps must be written immutably with actor identity and UTC timestamps.
- Every AI invocation must be persisted with prompt version, model identifier, inputs, raw outputs, validation results, token consumption, and latency to ensure full operational reproducibility.

Performance

- Standard non-AI API responses must complete within 500 ms at p95 under anticipated production load.
- Dashboard initial render must complete within 2.0 seconds at p95 over corporate network infrastructure.
- AI task generation must return within 20 seconds at p95, enforced by a hard 20-second timeout that automatically displays the manual task entry fallback.

Reliability & Availability

- The system must achieve 99.5% operational availability during core business hours, excluding planned maintenance windows.
- The application must remain 100% operational for all manual plan creation, tracking, and Information Hub browsing during total LLM provider downtime.

Scalability

- The architecture must support a minimum of 500 registered users and 100 concurrently active onboarding plans without infrastructure modification.

Usability & Accessibility

- Managers must be able to complete the generate-edit-approve onboarding workflow within a single screen without loss of form state.
- Joinees must be able to view their immediate weekly tasks and all overdue milestones on a single dashboard view without manual filtering.
- The web application must comply with WCAG 2.1 Level AA standards for keyboard navigation, contrast ratios, focus hierarchies, and screen-reader accessibility.
- Layouts must be fully responsive across Chrome, Edge, Firefox, and Safari down to 768px viewports.

Maintainability & Observability

- Prompt templates must be externalized in version-controlled repositories and validated against golden-output regression suites.
- Information Hub content must be editable via seed files and applied via database migration without rebuilding application artifacts.
- Telemetry must track API health, LLM latency, token expenditure, schema failure rates, and manager draft acceptance rates via structured JSON logs.

Cost & Localisation

- AI responses must be cached in Redis based on input context hashes to eliminate redundant token consumption on identical briefs.
- All UI strings must be decoupled into externalized resource bundles to facilitate future multi-language localization, with English supplied at launch.

---

7. Risks

ID	Risk	Impact	Mitigation
R-01	AI generates plausible but irrelevant or incorrect tasks, which a manager approves without review.	High	Enforce review via an interactive grid without bulk-approval buttons; require explicit assignment; warn on zero edits; track edit rates weekly; maintain golden prompt regression suites.
R-02	Third-party LLM provider experiences latency, rate limiting, or an outage, blocking plan assignment.	High	Implement a strict 20-second timeout with a single retry, immediately surfacing the manual task entry form on failure. Monitor provider health metrics.
R-03	Information Security rejects transmitting job and team context to an external LLM vendor late in development.	High	Secure written InfoSec boundary sign-off before Phase 1 build; restrict prompts strictly to job title and team; design the architecture to support a self-hosted LLM fallback.
R-04	Information Hub content becomes outdated (broken links, wrong emails, obsolete policies), misleading new hires.	High	Assign named content owners per category; record and display `` `last_reviewed_date` `` on every entry; implement quarterly review cadences; provide a "Report Error" link.
R-05	Managers bypass the system and revert to legacy ad-hoc emails and Slack messages.	High	Optimize the generate-edit-approve flow to under 20 minutes; pilot with 2–3 manager cohorts; make the dashboard the single organizational source for onboarding completion visibility.
R-06	Uncontrolled task regenerations cause LLM API token spend to exceed allocated operational budgets.	Medium	Cache responses by context hash; cap outputs at 15 tasks; rate-limit regenerations per plan; set daily spend alerts; monitor token cost per onboarded joinee.
R-07	LLM output frequently fails JSON schema validation, degrading the manager user experience.	Medium	Utilize provider-native structured outputs; validate server-side; execute one corrective retry; establish a <5% schema failure rate as an operational release gate.
R-08	A manager accesses, views, or modifies onboarding records belonging to another manager's direct reports.	High	Enforce server-side ownership checks on every API endpoint based on authenticated token claims; include negative authorization tests in CI/CD; log denied attempts.
R-09	Joinees mark compliance tasks and certifications as complete without	Medium	Require verification document uploads for mandatory certifications; display completion audit trails to managers to

ID	Risk	Impact	Mitigation
	performing the required work.		drive 1-on-1 check-ins.
R-10	Certification deadlines are omitted or entered incorrectly, creating compliance vulnerabilities.	Medium	Enforce mandatory flags and explicit date pickers for compliance certifications; display warnings for missing deadlines; automate overdue escalations.
R-11	Unapproved AI features are introduced into development, expanding regulatory and architectural scope.	Medium	Maintain the strict out-of-scope AI boundaries defined in this BRD; require formal Change Request approval for any model expansion.
R-12	Architectural coupling to a single commercial LLM vendor creates vendor lock-in.	Low	Wrap model invocations behind an internal service interface; standardize prompt schemas; validate compatibility with an alternate LLM provider prior to GA.
R-13	Low initial hiring intake leaves the platform underutilized, leading to stale reference content.	Low	Align system launch with known hiring cohorts; decouple Information Hub review schedules from active user counts.

8. Delivery Phases

Phase	Content	Exit Criteria
Phase 0 — Discovery and Documentation	Finalise this BRD and the PRD; produce the ERD and data dictionary; publish the OpenAPI contract; define the RBAC matrix; wireframe the nine screens; obtain the Information Security data-boundary decision; collect signed-off Information Hub content from IT and HR.	BRD and PRD approved by the decision audience; Information Security decision documented in writing; Information Hub content received and signed off; API contract published and agreed by backend and frontend.
Phase 1 — Foundation	Project scaffolding; PostgreSQL schema with Flyway migrations; OIDC authentication; MANAGER and JOINEE roles; server-side ownership enforcement; user creation and manager-to-joinee mapping; CI pipeline and deployment to a non-production environment.	Both roles can sign in and see an empty role-appropriate shell; a manager cannot access another manager's joinee, proven by an automated test; pipeline deploys green to the test environment.

Phase	Content	Exit Criteria
Phase 2 — Information Hub	`info_resource` table, seed loader and migration; category-based read-only Hub UI with accordion layout; keyword search; last-reviewed date displayed on every entry.	A joinee can find IT helpdesk contact, ticket-raising steps and access-request contacts without assistance; content is updatable by editing the seed file and redeploying, with no code change; IT and HR confirm displayed content is accurate.
Phase 3 — Manual Core	Plan creation with DRAFT and ACTIVE states; manual task CRUD with due-date calculation; certification CRUD with proof upload; connection CRUD; joinee weekly plan view and status updates; progress calculation; manager joinee list.	A manager can build, assign and track a complete onboarding plan end to end with no AI involved; a joinee can view everything assigned and update every status; progress figures reconcile against the underlying records.
Phase 4 — AI Task Generation	Prompt template store and output JSON schema; LLM client with retry, timeout and abstraction layer; server-side schema validation; editable draft grid; manual fallback path; AI run logging; Redis response caching; golden-output regression tests.	A manager can generate, edit and assign a plan from a free-text brief; a forced LLM outage still permits full manual assignment; every run is logged with acceptance and edit status; first-attempt schema failure rate below 5% across the test set.
Phase 5 — Notifications and Hardening	Scheduled due-date reminders and overdue escalations; audit trail completion; accessibility pass to WCAG 2.1 AA; performance testing against stated targets; security review; observability dashboards for AI cost, latency and acceptance.	Reminders and escalations verified in a staging run; accessibility audit passed; p95 latency targets met; security review closed with no high-severity findings.
Phase 6 — Pilot and Rollout	Pilot with two or three managers and their live joinees; collect feedback; tune prompts against real briefs; correct Information Hub content; then general availability with manager onboarding guidance.	Pilot joinees complete onboarding through the tool; AI acceptance rate above 70% on real usage; pilot managers confirm the flow is faster than their previous process; no open high-severity defects.

Note: Delivery phases are strictly sequential; Phase 0 through Phase 4 constitute the core system dependencies required before piloting in Phase 6, with Phase 5 delivering hardening, automated notification schedules, and accessibility compliance.

9. Open Questions

- What is the current baseline — how long does a manager spend today preparing an onboarding plan, and how many first-30-day tickets does IT receive per joinee?
- Has Information Security issued a position on sending job title and team to a third-party LLM provider, and is a self-hosted model required instead?
- Which LLM provider and model are approved, and what is the monthly budget cap?
- Who are the named content owners for each Information Hub category, and what review cadence will they commit to?
- Is the onboarding period fixed at 30 days, or must the plan duration be configurable per role?
- Should a joinee be able to see the Information Hub before their official start date, and if so how is their account provisioned early?
- Are certifications organisation-mandated and therefore standard across roles, or entirely manager-discretionary?
- Is certification proof upload actually required in v1, or is a self-declared status sufficient?
- What happens to an active plan when a joinee changes manager mid-onboarding?
- Can a joinee also be a manager — for example a senior hire who is themselves onboarding while owning direct reports?
- What is the expected hiring volume per month, which determines whether manual joinee creation is sustainable?
- Should the manager receive a digest email, or only exception-based overdue escalations?
- What are the data retention periods for completed plans, AI run logs and certification proof documents?
- Which deployment platform and environment tiers are available?
- Is there an existing internal design system or component library the frontend must adopt?

10. Future Scope (Deferred, Not Rejected)

Nothing in this section is committed, estimated, or approved for current release cycles.

Item	What it would unlock	What it would require
AI-generated introductory connections	Automated recommendation of relevant peers and stakeholders based on role.	Authoritative corporate directory integration, graph API access, and algorithmic fairness audits.
AI-written progress summaries	Executive narrative summaries of team onboarding progress on manager dashboards.	Prompt engineering for summarization, secondary review UI, and increased LLM operational budget.
RAG Joinee Assistant	Natural language conversational Q&A over internal documentation and handbooks.	Enterprise document index, vector database, embedding pipeline, and document ACL synchronization.
AI-suggested role certifications	Automated pre-population of mandatory compliance modules by job family.	Authoritative compliance certification matrix and formal legal/compliance sign-off.
HRIS Integration	Automatic user provisioning and reporting hierarchy synchronization upon hire.	HRIS API integration (e.g., Workday, BambooHR), webhook infrastructure, and sync error handling.
Slack & Calendar Integration	In-app meeting scheduling and interactive Slack notification actions.	Slack Bot application approval, OAuth scopes, Google/Outlook Calendar API integrations.
Reusable Plan Templates	Instant baseline plan instantiation for standardized recurring job titles.	Template management schema, versioning UI, and cross-team template curation.
Dedicated HR Administrator Role	Direct UI-driven management of Information Hub content and cross-company reporting.	Extended RBAC model, HR admin consoles, and tenant-wide permission boundaries.
Native Mobile Application	Native iOS/Android task management for field or non-desk employees.	Mobile engineering resources, app store deployment pipelines, and mobile MDM integration.
Cross-Org Onboarding Analytics	Benchmarking time-to-productivity and ramp-up bottlenecks across departments.	Data warehouse pipelines, historical cohort datasets, and BI reporting visualization tools.

11. AI Boundary Statement

Artificial intelligence is used for exactly one capability in this product: expanding a manager's short free-text brief into a structured draft task list. It is not used to select certifications, to recommend which people a joinee should meet, to summarise progress, or to generate or answer questions over Information Hub content. Those are either manager-entered or static seeded content by deliberate design.

The governing rule is that AI proposes and the manager approves. No AI-generated content becomes visible to a joinee without an explicit manager approval action, and no workflow in the product depends on the model being available. This boundary exists so that accountability for what a new joinee is told stays with a named human, so that the system remains predictable and auditable, and so that operating cost stays low and bounded. Any proposal to extend AI beyond this boundary requires a formal change request and re-approval by the decision audience.

---

12. AI Output Contract

The AI must return a JSON object containing a single `tasks` array. Each element must provide:

- `title` (string, 5 to 100 characters)
- `description` (string, up to 500 characters)
- `category` (enum: `TECHNICAL_SETUP`, `ACCESS_AND_TOOLS`, `COMPLIANCE`, `TEAM_AND_PROCESS`, `ROLE_RAMP_UP`, `ADMIN`)
- `week_no` (integer, 1 to 4)
- `day_offset` (integer, 0 to 30, expressed as days from the joinee's start date)
- `priority` (enum: `HIGH`, `MEDIUM`, `LOW`)
- `subtasks` (optional array of up to six short strings)

A maximum of 15 tasks may be returned in a single generation. The response is validated server-side against this schema before it reaches the manager. Malformed responses trigger exactly one retry with a corrective instruction; a second failure surfaces the manual entry form.

The AI never returns due dates — only day offsets — because concrete dates are derived at assignment time from the joinee's actual start date. This keeps generated content reusable and prevents date errors from propagating from the model.

---

13. Information Hub Content Structure

Information Hub content is stored in an `info_resource` table with the fields: `id`, `category`, `title`, `body_md` (Markdown), `contact_email`, `link_url`, `display_order`, `last_reviewed_date` and `is_active`. It is populated from a versioned seed file applied through a database migration, so that corrections require a content edit and redeploy rather than a code change.

The six launch categories are:

1. **IT and Access**: Covering the helpdesk contact, how to raise a support ticket, ticket SLA and VPN setup.
2. **Accounts and Tools**: Covering email, Slack, Jira, GitHub and who approves each access request.
3. **HR and Payroll**: Covering payroll contact, leave policy, holiday calendar and benefits.
4. **Workplace**: Covering office timings, seating, ID cards and reimbursement.
5. **Security and Compliance**: Covering the code of conduct, data handling policy and mandatory reading.
6. **FAQ**: A plain list of who to ask for what.

Each category has a named owner in IT or HR who is accountable for accuracy. The displayed `last_reviewed_date` makes staleness visible to joinees rather than hiding it.

---

14. Role and Permission Matrix

- **MANAGER** may:

    - Create and edit joinee records within their own reporting line.
    - Create, edit and delete plans, tasks, certifications and connections for their own joinees only.
    - Trigger AI task generation.
    - Approve and assign plans.
    - View progress for their own joinees.
    - Read the Information Hub.
- **MANAGER** may not:
    - Access any joinee outside their reporting line.
    - Edit Information Hub content directly in the application.
    - Modify a joinee's recorded task completion states.
- **JOINEE** may:
    - Read their own active plan, certifications and connections.
    - Update the status of their own tasks, certifications and connections.
    - Upload their own certification proof documents.
    - Add notes or blocker comments to their own tasks.
    - Read and search the Information Hub.
- **JOINEE** may not:
    - View any other joinee's onboarding data.
    - View draft plans prior to manager assignment.
    - Create, edit or delete assigned plan tasks, certifications or connections.
    - Invoke AI generation services.

All authorisation is enforced server-side on every request. Ownership is resolved from the authenticated token, never from an identifier supplied in the request body or query string.

---

15. Definition of MVP

The minimum viable product is Phases 0 through 4 inclusive: authentication with two roles and enforced ownership, the seeded Information Hub, full manual plan and task management, certifications, people to connect with, the joinee dashboard with progress and overdue visibility, the manager joinee list, and AI-assisted task drafting with a mandatory manager approval step and a working manual fallback.

Notifications, the full audit trail, accessibility certification and observability dashboards are Phase 5 and are not required for the pilot. The MVP is considered viable when one manager can onboard one real joinee end to end using only this tool, and the joinee needs no email or verbal instruction to know what to do, what to complete, whom to meet, or where to get IT and HR help.

---

16. Measuring AI Quality

AI quality is measured by manager behaviour, not by subjective output review. Three metrics are tracked per generation and reported weekly:

- **Acceptance Rate**: The share of generated tasks that survive to assignment without deletion, targeted above 70%.
- **Edit Rate**: The share of assigned tasks whose fields were modified before assignment, targeted below 30%, where a high figure signals the prompt is producing content that is close but consistently wrong in a specific dimension.
- **Regeneration Rate**: The number of times a manager re-runs generation for the same plan, where anything above one repeatedly indicates the first output is unusable.
- **Schema Validation Failure Rate**: Tracked separately on first attempts as a reliability signal, targeted below 5%.

A set of golden briefs with expected output characteristics is maintained as a regression suite, and every prompt version change is evaluated against it before release.

---

17. Out-of-Scope AI Uses and Why

Four AI capabilities were considered and deliberately excluded from v1:

1. **AI-recommended people to connect with**: Excluded because it requires an accurate org directory the organisation does not currently expose, and because an algorithm choosing whom a new employee should build relationships with raises fairness concerns that outweigh the convenience; the manager already knows the answer.
2. **AI-generated progress summaries**: Excluded because a completion percentage with an overdue count conveys the same information deterministically, at zero cost and with no risk of a misleading summary.
3. **AI assistant answering joinee questions over company documents**: Excluded because it requires a curated corpus, a vector store and a security review of which documents a joinee may see; the static Information Hub answers the common questions with guaranteed accuracy.
4. **AI-suggested certifications**: Excluded because certification requirements are compliance-driven, and a hallucinated or omitted mandatory certification carries genuine regulatory risk.

Each remains a candidate for a later release once v1 usage data justifies the additional cost and review burden.

---

18. Sign-Off

Role	Name	Signature	Date
Engineering Manager / Product Owner	[Name]	_____	[Date]
Head of HR	[Name]	_____	[Date]
IT Operations Lead	[Name]	_____	[Date]
Information Security Reviewer	[Name]	_____	[Date]